21.3 Triangles in Two and Three Dimensions

Not since the time of Euclid has the lowly triangle attracted as much attention as it does today in computer graphics. Triangles and *triangulation* (the decomposition, or approximation, of complicated geometrical objects using only triangles) are at the heart of practically every computer-generated image.

Three points, call them **a**, **b**, **c**, define a triangle. They are its *vertices*. If the points are two-dimensional, the triangle lies in the two-dimensional plane. If the points have higher dimensionality, then the triangle floats in the corresponding D-dimensional space (most commonly $D = 3$). For now, consider only the former case, with $D = 2$, so that **a** has coordinates (a_0, a_1), and similarly for **b** and **c**.

Area. The area $\mathcal{A}(\mathbf{abc})$ of the triangle $\triangle\mathbf{abc}$ can be written in a number of equivalent ways, including

$$
\begin{aligned}
2\mathcal{A}(\mathbf{abc}) &= \begin{vmatrix} a_0 & a_1 & 1 \\ b_0 & b_1 & 1 \\ c_0 & c_1 & 1 \end{vmatrix} \\
&= (\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a}) = (b_0 - a_0)(c_1 - a_1) - (b_1 - a_1)(c_0 - a_0) \\
&= (\mathbf{c}-\mathbf{b}) \times (\mathbf{a}-\mathbf{b}) = (c_0 - b_0)(a_1 - b_1) - (c_1 - b_1)(a_0 - b_0) \\
&= (\mathbf{a}-\mathbf{c}) \times (\mathbf{b}-\mathbf{c}) = (a_0 - c_0)(b_1 - c_1) - (a_1 - c_1)(b_0 - c_0)
\end{aligned}
\tag{21.3.1}
$$

Here $\times$ denotes the vector cross product, defined in two dimensions simply by

$$
\mathbf{A} \times \mathbf{B} = A_0 B_1 - B_1 A_0 \qquad \text{(two dimensions only)} \tag{21.3.2}
$$

Below, when we consider triangles in three dimensions, it will be the vector cross product forms in equation (21.3.1) that give a generalized formula for the area. Let us also note in passing that the formulas for area are separately *linear* in each of the six coordinates a_0, a_1, b_0, b_1, c_0, and c_1.

Equation (21.3.1) can yield a value that is positive, zero, or negative: The area is a *signed area*. By convention (embodied in equation 21.3.1), the area is positive if a traversal from **a** to **b** to **c** goes *counterclockwise (CCW)* around the triangle, and negative if it goes *clockwise (CW)*. The area is zero if and only if the three points are collinear, in which case the triangle is degenerate. (In the formulas that follow, we will generally assume the nondegenerate case.)

The absolute value $|\mathcal{A}|$ is the (unsigned) "area" of the triangle in the conventional geometrical sense. It can also be calculated directly from the side lengths d_{ab}, d_{bc}, and d_{ca} as follows:

$$
|\mathcal{A}| = \sqrt{s(s - d_{ab})(s - d_{bc})(s - d_{ca})} \tag{21.3.3}
$$

where s is half the perimeter,

$$
s \equiv \tfrac{1}{2}(d_{ab} + d_{bc} + d_{ca}) \tag{21.3.4}
$$

(Does it go without saying that you compute the side lengths by taking the coordinate differences and using the Pythagorean theorem?)

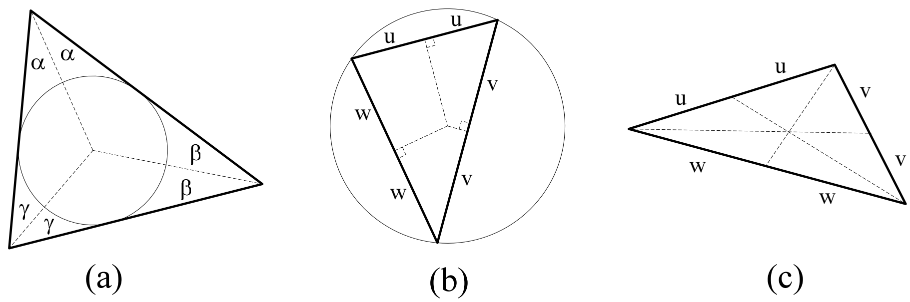

Figure 21.3.1. Three kinds of triangle centers. (a) Incircle and incenter; bisectors of the three vertex angles meet at the incenter. (b) Circumcircle and circumcenter; perpendicular bisectors of the edges meet at the circumcenter. (c) Centroid; lines from the edge midpoints to the opposite vertices meet at the centroid.

Related Circles. Every nondegenerate triangle has an *inscribed circle* or *incircle*, which is the largest circle that can be drawn inside the triangle. The incircle is tangent to all three sides of the triangle. Lines from its center, the *incenter*, to each vertex bisect the angle at that vertex (see Figure 21.3.1). If $\mathbf{q}$ is the incenter point, with coordinates (q_0, q_1), then its location is given by

$$q_i = \frac{1}{2s}(d_{bc}a_i + d_{ca}b_i + d_{ab}c_i) \qquad (i = 0, 1) \tag{21.3.5}$$

while its radius is given by

$$r_{\text{in}} = \left(\frac{(s - d_{ab})(s - d_{bc})(s - d_{ca})}{s}\right)^{1/2} \tag{21.3.6}$$

Every nondegenerate triangle also has a *circumscribed circle* or *circumcircle*, which is the unique circle that goes through its three vertices. Suppose $\mathbf{Q}$ is the *circumcenter* point, with coordinates (Q_0, Q_1). Let $[ba]_0$ and $[ba]_1$ denote the coordinate differences $b_0 - a_0$ and $b_1 - a_1$, respectively; and similarly for $[ca]_0$ and $[ca]_1$. Then, in 2×2 determinant form,

$$\begin{aligned}
Q_0 &= a_0 + \frac{1}{2}\begin{vmatrix} ([ba]_0)^2 + ([ba]_1)^2 & [ba]_1 \\ ([ca]_0)^2 + ([ca]_1)^2 & [ca]_1 \end{vmatrix} \Big/ \begin{vmatrix} [ba]_0 & [ba]_1 \\ [ca]_0 & [ca]_1 \end{vmatrix} \\
Q_1 &= a_1 + \frac{1}{2}\begin{vmatrix} [ba]_0 & ([ba]_0)^2 + ([ba]_1)^2 \\ [ca]_0 & ([ca]_0)^2 + ([ca]_1)^2 \end{vmatrix} \Big/ \begin{vmatrix} [ba]_0 & [ba]_1 \\ [ca]_0 & [ca]_1 \end{vmatrix}
\end{aligned} \tag{21.3.7}$$

The circumcenter is, by definition, the same distance from all three vertices. Therefore the radius of the circumcircle is

$$r_{\text{circum}} = \sqrt{(Q_0 - a_0)^2 + (Q_1 - a_1)^2} \tag{21.3.8}$$

where Q_0 and Q_1 are given above. (Obviously you can save the semi-final results in equation 21.3.7 for this computation, before adding a_0 or a_1.)

Later, in §21.6, we will be calculating a lot of circumcircles. We use the following simple definition of a structure `Circle`, and a routine `circumcircle()` that directly implements equations (21.3.7) and (21.3.8).

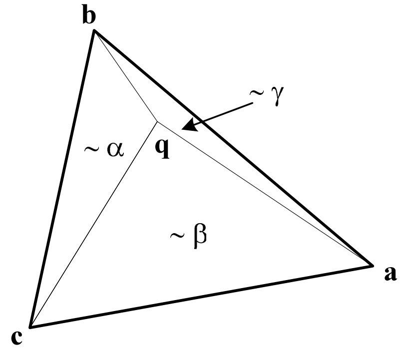

Figure 21.3.2. Any point in the plane **q** can be expressed as a linear combination of a triangle's three vertices. The coefficients (α, β, γ), called barycentric coordinates, sum to 1 and are proportional to the areas of $\triangle\mathbf{qbc}$, $\triangle\mathbf{qca}$, and $\triangle\mathbf{qab}$, respectively.

circumcircle.h

```
struct Circle {
    Point<2> center;
    Doub radius;
    Circle(const Point<2> &cen, Doub rad) : center(cen), radius(rad) {}
};

Circle circumcircle(Point<2> a, Point<2> b, Point<2> c) {
    Doub a0,a1,c0,c1,det,asq,csq,ctr0,ctr1,rad2;
    a0 = a.x[0] - b.x[0]; a1 = a.x[1] - b.x[1];
    c0 = c.x[0] - b.x[0]; c1 = c.x[1] - b.x[1];
    det = a0*c1 - c0*a1;
    if (det == 0.0) throw("no circle thru colinear points");
    det = 0.5/det;
    asq = a0*a0 + a1*a1;
    csq = c0*c0 + c1*c1;
    ctr0 = det*(asq*c1 - csq*a1);
    ctr1 = det*(csq*a0 - asq*c0);
    rad2 = ctr0*ctr0 + ctr1*ctr1;
    return Circle(Point<2>(ctr0 + b.x[0], ctr1 + b.x[1]), sqrt(rad2));
}
```

Centroid and Barycentric Coordinates. Distinct from both its incenter and its circumcenter is a triangle's centroid, or center of gravity, **M**. This point lies at the intersections of the lines drawn from each vertex to the midpoint of the opposite side. Its coordinates are simply the means of the coordinates of the vertices,

$$M_i = \tfrac{1}{3}(a_i + b_i + c_i) \qquad (i = 0, 1) \tag{21.3.9}$$

The centroid is also the point **M** where the areas $\mathcal{A}(\mathbf{abM})$, $\mathcal{A}(\mathbf{bcM})$, and $\mathcal{A}(\mathbf{caM})$ are all equal. In §21.7 we will be using a triangular mesh to interpolate a function. The significance of the centroid is that it is the point where a linearly interpolated function takes on the value that is the mean of the function values at the three vertices.

In fact, generalizing the idea of the centroid, any point **q** in the plane can be written as a linear combination of the three vertices **a**, **b**, **c**, with coefficients that sum to unity. These coefficients are called **q**'s *barycentric coordinates* and can be intuitively expressed in terms of the area formulas for triangles (see Figure 21.3.2). The equations are

$$
\begin{aligned}
\mathbf{q} &= \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c} \\
\alpha &= \mathcal{A}(\mathbf{bcq})/\mathcal{A}(\mathbf{abc}) \\
\beta &= \mathcal{A}(\mathbf{caq})/\mathcal{A}(\mathbf{abc}) \\
\gamma &= \mathcal{A}(\mathbf{abq})/\mathcal{A}(\mathbf{abc})
\end{aligned}
\tag{21.3.10}
$$

with, by construction,

$$\alpha + \beta + \gamma = 1 \tag{21.3.11}$$

The first line in equation (21.3.10) is thus equivalent to

$$\mathbf{q} = \mathbf{c} + \alpha(\mathbf{a} - \mathbf{c}) + \beta(\mathbf{b} - \mathbf{c}) \tag{21.3.12}$$

This can be viewed as the equation for a coordinate transformation, one that transforms from (α, β) coordinates to (q_0, q_1) coordinates. Evidently, since it is linear, its inverse — the formulas for α and β in equation (21.3.10) — must also be linear. But we knew this already, having remarked on the fact that the area formulas (21.3.1) are linear in all their coordinates, so linear in q_0 and q_1 in particular. Barycentric coordinates generalize to triangles in three or more dimensions in a useful way, as we will see below.

Note that α, β, or γ go to 1 as the point $\mathbf{q}$ approaches $\mathbf{a}$, $\mathbf{b}$, or $\mathbf{c}$, respectively; and that along any edge of the triangle (say $\overline{ab}$) the coefficient of the opposite vertex (here, γ) vanishes. The point $\mathbf{q}$ is inside the triangle $\triangle\mathbf{abc}$ if and only if α, β, and γ are all positive. In fact, this is a good way to test for a point's "insideness" in a triangle. (You can of course omit calculating the denominator area in this application.)

Barycentric coordinates are also useful when you want to pick a uniformly random point $\mathbf{q}$ inside $\triangle\mathbf{abc}$: First pick α and β as each uniformly random in $(0, 1)$. Next, if $\alpha + \beta > 1$, modify them both by $\alpha \leftarrow 1 - \alpha$ and $\beta \leftarrow 1 - \beta$. Finally, apply equation (21.3.12). The idea is that the first choice of α and β is random in the parallelogram spanned by two sides of the triangle; then, if it is on the wrong side of the diagonal, we move it to the correct side by a reflection.

21.3.1 Triangles in Three Dimensions

Our favorite triangle is still defined by the three points $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, but these are now points in three dimensions, with coordinates (e.g., for $\mathbf{a}$) (a_0, a_1, a_2). The generalization of the signed area $\mathcal{A}$ (equation 21.3.1) is now a vector area $\vec{\mathcal{A}}$ whose direction is normal to the plane of the triangle and whose length is the area of the triangle. It is most easily written using a vector cross product, defined in three dimensions by

$$
\begin{aligned}
\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{\mathbf{e}}_0 & \hat{\mathbf{e}}_1 & \hat{\mathbf{e}}_2 \\ A_0 & A_1 & A_2 \\ B_0 & B_1 & B_2 \end{vmatrix} \\
&= (A_1 B_2 - A_2 B_1)\hat{\mathbf{e}}_0 + (A_2 B_0 - A_0 B_2)\hat{\mathbf{e}}_1 + (A_0 B_1 - A_1 B_0)\hat{\mathbf{e}}_2
\end{aligned}
\tag{21.3.13}
$$

where $\hat{\mathbf{e}}_0$, $\hat{\mathbf{e}}_1$, and $\hat{\mathbf{e}}_2$ are respectively the unit vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. Then we have (cf. equation 21.3.1)

$$\begin{aligned} 2\,\vec{\mathcal{A}}(\mathbf{abc}) &= (\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a}) \\ &= (\mathbf{c}-\mathbf{b}) \times (\mathbf{a}-\mathbf{b}) \\ &= (\mathbf{a}-\mathbf{c}) \times (\mathbf{b}-\mathbf{c}) \end{aligned} \tag{21.3.14}$$

You calculate the positive scalar area $\mathcal{A} \equiv |\vec{\mathcal{A}}|$ by the usual square-root sum of the squares of $\vec{\mathcal{A}}$'s three components; or you can instead use equation (21.3.3), with $d_{ab} = |\mathbf{a}-\mathbf{b}|$, etc.

Plane Defined by Triangle. A point $\mathbf{q}$ lies in the plane defined by $\triangle\mathbf{abc}$ if and only if the volume of the tetrahedron $\mathbf{abcq}$ is zero. The tetrahedral volume, in general, is given by

$$\begin{aligned} 6\,\mathcal{V} &= \begin{vmatrix} a_0 & a_1 & a_2 & 1 \\ b_0 & b_1 & b_2 & 1 \\ c_0 & c_1 & c_2 & 1 \\ q_0 & q_1 & q_2 & 1 \end{vmatrix} \\ &= (\mathbf{b}-\mathbf{a}) \cdot [(\mathbf{c}-\mathbf{a}) \times (\mathbf{q}-\mathbf{a})] \\ &= (\mathbf{c}-\mathbf{a}) \cdot [(\mathbf{q}-\mathbf{a}) \times (\mathbf{b}-\mathbf{a})] \\ &= (\mathbf{q}-\mathbf{a}) \cdot [(\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a})] \end{aligned} \tag{21.3.15}$$

where "·" signifies vector dot product. You can also cyclically permute $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ in the above equation, for a seemingly infinite number of variations of the same formula!

The volume $\mathcal{V}$ is signed and is positive if $\triangle\mathbf{abc}$ is counterclockwise when viewed from outside (side away from $\mathbf{q}$), that is, the right-hand rule gives an outward-pointing normal.

The last form in equation (21.3.15) is particularly nice, because setting it to zero gives the equation satisfied by any point $\mathbf{q}$ in the plane defined by $\triangle\mathbf{abc}$:

$$\mathbf{q} \cdot \mathbf{N} = D \tag{21.3.16}$$

with

$$\begin{aligned} \mathbf{N} &= (\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a}) \quad \text{(or cyclic permutation of } \mathbf{a}, \mathbf{b}, \mathbf{c}) \\ D &= \mathbf{a} \cdot \mathbf{N} \quad \text{(or, for that matter)} \quad = \mathbf{b} \cdot \mathbf{N} = \mathbf{c} \cdot \mathbf{N} \end{aligned} \tag{21.3.17}$$

We could also divide equation (21.3.16) by $|\mathbf{N}|$, in which case the vector on the left will be $\widehat{\mathbf{N}} = \mathbf{N}/|\mathbf{N}|$, the unit vector normal to the plane, and $\widehat{D} = D/|\mathbf{N}|$ will be the plane's distance from the origin.

With the same machinery, we can readily project any point $\mathbf{p}$ into a new point $\mathbf{b}'$ that lies in the plane of $\triangle\mathbf{abc}$:

$$\mathbf{p} \longrightarrow \mathbf{p}' = \mathbf{p} + \frac{[(\mathbf{a}-\mathbf{p}) \cdot \mathbf{N}]\mathbf{N}}{|\mathbf{N}|^2} \tag{21.3.18}$$

where $\mathbf{N}$ is as above. For $\mathbf{a}$ in this formula, you can substitute $\mathbf{b}$, $\mathbf{c}$, or any other point in the plane.

We can project one triangle into the plane defined by another triangle by projecting its three points in turn. (This is a very common operation in rendering a triangulated three-dimensional model in the two-dimensional "camera plane" of your computer's screen.)

Barycentric Coordinates. Barycentric coordinates are valid in three dimensions for points $\mathbf{q}$ in the triangle's plane, and equation (21.3.10), in particular, still holds. To compute (α, β), one can in principle calculate the various $\mathcal{A}$'s from (21.3.14), but an easier equivalent calculation is

$$\alpha = \frac{\mathbf{b}'^2(\mathbf{a}' \cdot \mathbf{q}') - (\mathbf{a}' \cdot \mathbf{b}')(\mathbf{b}' \cdot \mathbf{q}')}{\mathbf{a}'^2\mathbf{b}'^2 - (\mathbf{a}' \cdot \mathbf{b}')^2}$$
$$\beta = \frac{\mathbf{a}'^2(\mathbf{b}' \cdot \mathbf{q}') - (\mathbf{a}' \cdot \mathbf{b}')(\mathbf{a}' \cdot \mathbf{q}')}{\mathbf{a}'^2\mathbf{b}'^2 - (\mathbf{a}' \cdot \mathbf{b}')^2} \tag{21.3.19}$$

(compute identical denominators only once) where

$$\mathbf{a}' \equiv \mathbf{a} - \mathbf{c}, \qquad \mathbf{b}' \equiv \mathbf{b} - \mathbf{c}, \qquad \mathbf{q}' \equiv \mathbf{q} - \mathbf{c} \tag{21.3.20}$$

By the way, if $\mathbf{q}$ is not in the plane of $\triangle\mathbf{abc}$, you can still use equation (21.3.19). In that case, you get the (α, β) coordinates of the projected point in the plane. Also, notice what happens in the special case that $\triangle\mathbf{abc}$ is a right triangle, with right vertex $\mathbf{c}$, and with sides $\overline{ac}$ and $\overline{bc}$ of unit length, i.e., $d_{ac} = d_{bc} = 1$. Then the coordinate transformations, in both directions, are simply

$$\begin{aligned} \mathbf{q} &= \mathbf{c} + \alpha(\mathbf{a} - \mathbf{c}) + \beta(\mathbf{b} - \mathbf{c}) \\ [\alpha, \beta] &= [(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{q} - \mathbf{c}), (\mathbf{b} - \mathbf{c}) \cdot (\mathbf{q} - \mathbf{c})] \end{aligned} \tag{21.3.21}$$

In other words, we project into an orthonormal coordinate system in the plane by a simple change of origin (to $\mathbf{c}$) and dot products with the two "axes" $\mathbf{a} - \mathbf{c}$ and $\mathbf{b} - \mathbf{c}$.

Frequently, barycentric coordinates are the coordinates of choice for operations in a plane in three dimensions that is (or can be) specified by a triangle. A trivial example is that we can test whether a projected point $\mathbf{p}'$ is inside or outside of $\triangle\mathbf{abc}$ by using equation (21.3.19) (or, if applicable, 21.3.21) to get α and β, and then checking that α, β, and $\gamma = 1 - \alpha - \beta$ are all positive.

Angle Between Two Triangles. The dihedral angle between two triangles (with a common edge, say) is the same as the angle between the normal vectors of the two triangles. The normal vectors are given by the vector area formula (21.3.14). The angle is best computed using equation (21.4.13), in the next section.

CITED REFERENCES AND FURTHER READING:

Bowyer, A. and Woodwark, J. 1983, *A Programmer's Geometry* (London: Butterworths), Chapter 4.

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López-López, F.J. 1992, "Triangles Revisited," in *Graphics Gems III*, Kirk, D., ed. (Cambridge, MA: Academic Press).

Glassner, A.S. 1990, "Useful 3D Geometry," in *Graphics Gems*, Glassner, A.S., ed. (San Diego: Academic Press).